Spread Locator

A Statistical Distribution Analysis Model

Part A — Theoretical Foundation

Duration: 6 Hours | Type: Theory + Practical
Domain: E-commerce Transaction Analytics

"*Quality is our Motto.*"
Shaping "skills" for "scaling" higher...!!!

Q1. What is Statistical Distribution?

A statistical (probability) distribution describes how the values of a random variable are spread out — which values are likely, which are rare, and how probability is allocated across the range of possible outcomes. It is defined mathematically by a function (a PMF for discrete variables or a PDF for continuous variables) together with parameters that control its shape, such as the mean, variance, or shape/scale parameters.

Distributions let us summarise raw data with a small number of parameters, compare a dataset against a theoretical model, and make probabilistic predictions (e.g. "What is the probability a transaction exceeds ₹5000?"). In the Spread Locator project, we fit several standard distributions (Bernoulli, Binomial, Poisson, Log-Normal, Power-Law) to real transaction data to understand customer purchase behaviour.

> *Formally: for a random variable X, the distribution specifies P(X = x) (discrete) or the density f(x) such that P(a ≤ X ≤ b) = ∫[a,b] f(x) dx (continuous).*

Q2. What is a Q-Q Plot and why is it used?

A Quantile-Quantile (Q-Q) Plot is a graphical technique used to check whether a dataset follows a particular theoretical distribution — most commonly the Normal distribution. It plots the quantiles of the observed (sample) data on one axis against the quantiles of the theoretical distribution on the other axis.

- If the points fall approximately along the 45-degree reference line, the data closely follows the theoretical distribution.
- Curvature or an S-shape indicates skewness (right/left) or heavier/lighter tails than the reference distribution.
- Points that bow upward at both ends indicate positive skew (right-skewed data) — exactly what we observe for raw transaction amounts.

Q-Q plots are preferred over histograms for normality checks because they are far more sensitive to departures in the tails, which is where financial/transactional data usually deviates most from Normality. In this project, the Q-Q plot is used to confirm that raw transaction amounts are non-normal, while log-transformed amounts are approximately normal.

Q3. Difference between Discrete and Continuous Distributions

Aspect	Discrete Distribution	Continuous Distribution
Values taken	Countable set (0,1,2,...)	Any real value in an interval
Described by	Probability Mass Function (PMF)	Probability Density Function (PDF)

Probability at a point	P(X = x) can be > 0	P(X = x) = 0; only P(a≤X≤b) is meaningful
Examples	Bernoulli, Binomial, Poisson	Normal, Log-Normal, Power-Law
Project usage	transaction_status, transaction_count	transaction_amount

Q4. What is Bernoulli Distribution?

The Bernoulli distribution models a single trial that has exactly two possible outcomes: "success" (1) with probability p, or "failure" (0) with probability (1-p). It is the simplest discrete distribution and forms the building block of the Binomial distribution (n independent Bernoulli trials).

> *PMF: $P(X = x) = p^{x} (1-p)^{(1-x)}$, for $x \in \{0, 1\}$*
> *Mean: $E[X] = p$ Variance: $Var(X) = p(1-p)$*

Project application: In the Spread Locator dataset, the field transaction_status (Success/Fail) is a direct Bernoulli variable. We estimated $\hat{p} \approx 0.445$, meaning about 44.5% of transactions succeed.

Q5. What is Binomial Distribution?

The Binomial distribution gives the probability of obtaining exactly k successes in n independent, identical Bernoulli(p) trials. It is used when we count "successes" out of a fixed, known number of attempts.

> *PMF: $P(X = k) = C(n, k)\, p^{k} (1-p)^{(n-k)}$, $k = 0,1,\ldots,n$*
> *Mean: $E[X] = np$ Variance: $Var(X) = np(1-p)$*

Project application: We modelled the weekly transaction_count of a customer as Binomial($n=9$, $\hat{p}\approx 0.317$), using the maximum observed weekly count as n. A chi-square goodness-of-fit test rejected this model ($p \approx 0.00004$) because mean ≈ variance in the data (2.85 vs 3.23), which is the signature of a Poisson process rather than a Binomial one — an important, data-driven finding of this project.

Q6. Explain Log-Normal Distribution

A random variable X is Log-Normally distributed if ln(X) follows a Normal distribution. It is the natural model for strictly positive quantities that are the product of many small independent multiplicative effects — such as income, transaction sizes, city populations, and stock prices — which is why it fits transaction amounts so well.

> *PDF: $f(x) = 1 / (x \sigma \sqrt{2\pi}) \cdot \exp(-((\ln x - \mu)^2) / (2\sigma^2))$, $x > 0$*
> *Mean of X: $\exp(\mu + \sigma^2/2)$ Median of X: $\exp(\mu)$*

Project application: transaction_amount was fitted with $\mu \approx 8.00$, $\sigma \approx 0.47$. A Kolmogorov-Smirnov test could not reject the Log-Normal hypothesis ($p \approx 0.90$), while it strongly rejected a plain Normal fit ($p \approx 0.0003$) — confirming Log-Normal as the best model for spending amounts, which are right-skewed and cannot be negative.

Q7. Explain Power Law Distribution

A Power-Law (Pareto-type) distribution describes quantities where the probability of very large values decreases polynomially rather than exponentially, producing a long, "heavy" tail — the well-known 80/20 phenomenon (city sizes, wealth, word frequencies, social-network degree distributions).

> *PDF (continuous, $x \geq x\_min$): $f(x) = ((\alpha-1)/x\_min) \cdot (x/x\_min)^{-\alpha}$*
> *On a log-log plot, a true power law appears as a straight line with slope $-(\alpha-1)$.*

Project application: We tested the tail of transaction_amount ($x \geq$ median) using the Hill/MLE estimator and obtained $\alpha \approx 3.92$ with a relatively large standard error, and the log-log CCDF was visibly curved rather than a straight line. This tells us the tail of spending amounts is not a genuine power law — Log-Normal remains the better overall model.

Q8. What is Box-Cox Transform?

The Box-Cox transform is a family of power transformations used to convert a non-normal, often skewed, positive random variable into a shape that is approximately Normal, stabilising variance and improving the validity of statistical techniques (t-tests, regression, control charts) that assume normality.

> *$y(\lambda) = (x^{\lambda} - 1)/\lambda$, for $\lambda \neq 0$ $y(\lambda) = \ln(x)$, for $\lambda = 0$*

The optimal λ is chosen (typically via maximum likelihood) to make the transformed data as close to Normal as possible. Project application: for transaction_amount, the optimal $\lambda \approx -0.18$ — close to 0, the value at which Box-Cox becomes a pure log transform. After transformation, skewness collapsed from 3.73 to virtually 0, confirming Log-Normal/log-scale as the right lens for this data.

Q9. Explain Poisson Distribution with an example

The Poisson distribution models the number of independent events occurring in a fixed interval of time or space, given a constant average rate λ, when events happen one at a time and are memoryless (the time since the last event doesn't affect the next). It is the limiting case of the Binomial distribution when n is large, p is small, and $np = \lambda$ is moderate.

> *PMF: $P(X = k) = (\lambda^k e^{-\lambda}) / k!$, $k = 0,1,2,\ldots$*
> *Mean: $E[X] = \lambda$ Variance: $Var(X) = \lambda$ (mean = variance is the key diagnostic)*

Example: If a helpdesk receives an average of 5 calls per hour ($\lambda=5$), the Poisson distribution gives the probability of receiving exactly k calls in the next hour. Project application: the number of transactions per day on the e-commerce platform was modelled as Poisson($\lambda \approx 7.10$); a chi-square test could not reject this fit ($p \approx 0.23$). Similarly, the weekly transaction_count per customer fit Poisson($\lambda \approx 2.85$) far better than Binomial ($p \approx 0.71$ vs $p \approx 0.00004$).

Q10. What is Z-score Probability?

A Z-score standardises an observation by expressing how many standard deviations it lies above or below the mean of its distribution. It allows values from any Normal (or approximately Normal) distribution to be compared on a common scale and converted into probabilities using the standard Normal distribution N(0,1).

> *$Z = (X - \mu) / \sigma$*
> *$P(X > x) = 1 - \Phi(Z)$, where Φ is the standard Normal CDF*

Project application: For a transaction of ₹5000 (mean $\mu \approx$ ₹3365, std $\sigma \approx$ ₹1986), $Z \approx 0.82$. A plain Normal approximation estimates $P(X>5000) \approx 20.5\%$, but because the true distribution is Log-Normal (right-skewed), the actual empirical probability is only ≈11.4% — the Log-Normal-based estimate (≈13.8%) is far closer to reality. This demonstrates why the underlying distribution shape must be checked before trusting Z-score-based probabilities.

Q11. Differentiate Probability Density Function (PDF) and Cumulative Distribution Function (CDF)

Aspect	PDF	CDF
Definition	Relative likelihood density of X near a point x	Probability that X is less than or equal to x
Notation	$f(x)$	$F(x) = P(X \leq x)$
Relationship	$f(x) = dF(x)/dx$ (derivative of CDF)	$F(x) = \int f(t)\,dt$, from $-\infty$ to x
Range of values	$f(x) \geq 0$, area under curve = 1	Monotonically increasing from 0 to 1
Shape (this project)	Bell-shaped/right-skewed hump peaking near the mode of transaction amounts	S-shaped curve rising from 0 to 1 across the amount range

Project application: For transaction_amount, the fitted Log-Normal PDF shows the density peaking around ₹2500-3000 and tapering off with a long right tail; the corresponding CDF was used to directly read off $P(X \leq 5000) \approx 86\%$ (so $P(X > 5000) \approx 14\%$), matching the Log-Normal probability computed via the Z-score/CDF method.